

DEFENSE MEDICAL COLLEGE

Progressive Test I in Histology for Year I Medical Students

Time allowed: 2:00 hours.

Date: March 7, 2014

Name: _____

PART I. For each of the following **Multiple Choice Questions** choose the one **BEST** appropriate answer and write your answer in **BLOCK LETTERS** in the appropriate space located against the corresponding question number.

_____1. **1000 micrometers** is equivalent to:

- a) 1 centimeter. b) 100 nanometers. c) 10 millimeters. d) 1 millimeter. e) 0.1 millimeter

_____2. The stain of choice to demonstrate **glycogen** and/or **mucus** in cells is:

- a. hematoxylin. b. eosin. c. lipofuchsin. d. hemosiderin. e. periodic acid Schiff (PAS).

_____3. Which of the following organelles is the site of **steroid hormone synthesis**?

- a) Golgi complex b) mitochondria c) lysosomes d) RER e) SER

_____4. Which one is **not** the function of the **peroxisomes**?

- a. Regulation of H₂O₂ within cells. b. Gluconeogenesis. c. β -oxidation of long chain fatty acids.
d. Formation of bile salts. e. cellular attachment to extracellular matrix components.

_____5. Which of the following pairs of functions is most closely associated with the **Golgi complex**?

- a) energy metabolism and glycogen synthesis b) energy metabolism and lipid metabolism
c) glycosylation and sulfation of secretory products d) phagocytosis and receptor recycling
e) glycogen synthesis and packaging of secretions.

_____6. How many somatic chromosomes are in a normal **human karyotype**?

- a) 22 b) 23 c) 24 d) 44 e) 46

_____7. Which of the following is **NOT** considered an inclusion?

- a. Pigment b. Glycogen c. Lipid d. Secretory granules e. Mitochondria

_____8. Which of the following is **NOT** the characteristic feature of the **centrioles**?

- a. are located at the base of cilia and flagella. b. possess low resistance junction.
c. are involved in cell division. d. are not membranous organelles. e. are composed of microtubules

_____9. With regard to the **cell membrane**, which statement is **false**?

- a) Lipid bilayer is freely permeable to charged ions. b) Cholesterol is located in both leaflets. c)
Peripheral proteins are found in the inner leaflet only (do not extend into the lipid bilayer).
d) Glycolipids, which form part of glycocalyx, are found in the outer leaflet only. e) None of the above.

—————10. Which of the following statements is **NOT** the **function of the membrane proteins**?

- a) Transport molecules to or out of the cells.
- b) Act as receptors for chemical signaling between cells.
- c) Attach cytoskeletal filaments to cell membrane.
- d) Attach cells to extracellular matrix.
- e) None of these

—————11. With regard to the **ribosomes**, select the **incorrect statement**.

- a) Ribosomes are sites where mRNA is translated into protein.
- b) Ribosomes are sites of protein synthesis and are intensely basophilic.
- c) Ribosomes are sites where rRNA is manufactured and are involved in the breakdown of hydrogen peroxide.
- d) Ribosomes are non-membranous organelles which read mRNA, recruit tRNA molecules to fetch amino acids, and assemble the amino acids in the proper order.
- e) Ribosomes are assembled in the nucleoli.

—————12. Which of the following contain the **most lysosomes**?

- a) fibroblasts
- b) lymphocytes
- c) macrophages
- d) mast cells
- e) plasma cells

—————13. The **Barr body**,

- a) its DNA replicates during every other cell cycle in normal males.
- b) in normal females, it is a member of homologous chromosome pair.
- c) is composed of euchromatin during the mitotic phase of a cell cycle.
- d) attaches to the inner surface of the plasma membrane during interphase.
- e) occurs when homologous chromosomes fail to separate during an anaphase.

—————14. Which of the following cells is **multinucleated**?

- a. Hepatocyte
- b. neutrophil
- c. parietal cell
- d. megakaryocyte
- e. None of these

—————15. With regard to the **cellular organelles**, which statement is **true**?

- a) Clathrin-coated vesicles are associated with constitutive protein transport.
- b) Lysosomal enzymes have optimal activity at basic pH for the synthesis of rRNA.
- c) Golgi apparatus possess heterochromatin which is transcriptionally active.
- d) The nucleolus is a well-defined, membrane-bounded acidophilic nuclear inclusion.
- e) Hemosiderin is found in phagocytic cells of the liver, bone marrow and spleen.

—————16. Concerning the **morphology of a cell**, select the **correct statement**.

- a) Microtubules are more stable than microfilaments.
- b) G₀ phase is the longest and most variable cell-cycle phase in continuously dividing cells.
- c) Lipofuscin is found in non-dividing cells (cardiac muscle cells, neurons) and hepatocytes.
- d) Ion-transporting cells contain abundant basal mitochondria between infoldings of the basal plasma membrane.
- e) The nucleolus disperses during the anaphase of cell division but reappears in the metaphase stage of mitosis.

—————17. Which of the following **intermediate filaments** is a marker for **connective tissue tumors**?

- a. Keratin
- b. Desmin
- c. Vimentin
- d. Neurofilaments
- e. Laminin

—————18. With regard to the **epithelial tissue**, select the **incorrect statement**.

- a) The lining of the urinary bladder is transitional epithelium (urothelium).
- b) Holocrine secretion is the typical mode of secretion of DNES cells.
- c) An example of epithelioid tissue is the parenchyma of the adrenal gland.
- d) Endothelium is simple squamous epithelium that lines the vascular system.
- e) Merocrine secretion is characterized by discharge of a secretion product with no loss of cytoplasm or membrane.

————19. **Stereocilia**

- a) are structurally similar to true cilia
- b) contain axonemes
- c) are underlain by a basal body
- d) contain actin filaments in their core
- e) contain nine microtubule triplets with paired central sheath.

————20. With regard to the **epithelial tissue**, which statement is **true**?

- a) Desmosomes permit the passage of ions between adjacent cells.
- b) Microvilli contain sheaf of microtubules with a 9+2 axoneme pattern.
- c) In pseudostratified columnar epithelium all cells touch the basal lamina.
- d) Arteriovenous anastomoses provide nutrition to the highly cohesive and polarized rows of cells.
- e) The major constituents of epithelia is abundant extracellular matrix with amorphous ground substance.

————21. **Mesothelium lines:**

- a. **the respiratory and the urinary tracts.**
- b. **the peritoneal, pleural and pericardial cavities.**
- c. **the nasal and the oral cavities.**
- d. **the gastrointestinal and reproductive tracts**
- e. **the central nervous system and the muscular system**

————22. With regard to the **epithelial tissue**, which statement is **false**?

- a) Stratified squamous epithelium forms the epidermis and the esophagus.
- b) Simple columnar epithelium lines most of the gastrointestinal tract.
- c) Simple squamous epithelium forms the alveoli in the lung.
- d) Conjunctiva is lined by stratified columnar epithelium.
- e) None of the above.

————23. Which surface is **NOT lined by epithelial tissue**?

- a. surface of skin.
- b. corneal surface of the eye.
- c. luminal surface of intestinal tract.
- d. luminal surface of trachea and alveoli of lungs.
- e. surfaces of bones and muscles.

————24. With regard to the **connective tissue**, which statement is **false**?

- a) Mucous connective tissue lacks connective tissue cells.
- b) Plasma cells characteristically synthesize and secrete immunoglobulins.
- c) Fibroblast produces and maintains all extracellular matrix components of connective tissue.
- d) Mast cells do not secrete catalase.
- e) White adipose tissue tends to increase as humans age.

————25. With regard to the **connective tissue**, which statement is **false**?

- a) Dense irregular connective tissue makes up the dermis of the skin.
- b) Collagen is characterized by high tensile strength and flexibility.
- c) In ordinary connective tissue, the predominant component of ground substance is water.
- d) Articular cartilage lacks a perichondrium.
- e) None of the above.

————26. Select the **correct statement about connective tissue**.

- a. Dense connective tissue is more abundant than loose connective tissue.
- b. Elastic cartilage is the most abundant type of cartilage.
- c. Proteoglycans are glycosaminoglycans in a hyaline cartilage.
- d. Fibrocartilage is present in the temporomandibular joint.
- e. Epiglottis is made up of hyaline cartilage.

————27. Which of the following statements is **false** regarding the **connective** and **supporting tissue**?

- a) Sharpey's fibers are the spicules seen within spongy bone.
- b) Howship's lacuna is a space seen underneath an osteoclast.
- c) Osteoid is unmineralized bone matrix.
- d) Dense regular connective tissue may be found in tendons.
- e) None of the above.

_____28. With regard to **fibroblasts**, which statement is **false**? **Fibroblasts**:

- a. produce collagen.
- b. are tissue macrophages.
- c. are connective tissue cells derived from mesenchyme
- d. are routinely seen in loose connective tissue
- e. produce the ground substance in connective tissue.

_____29. Beauty treatment for the reduction of wrinkles is the injection of hyaluronic acid into the wrinkle.

What is **hyaluronic acid**?

- a. Dermatan sulfate
- b. Proteoglycan
- c. Glycosaminoglycan
- d. Chondroitin sulfate
- e. Keratan sulfate

_____30. Which of the following is **NOT primarily composed of connective tissue**?

- a. Blood
- b. Bone
- c. Tendon
- d. Intervertebral disc
- e. Myometrium

_____31. Which type of cartilage forms the **malleus (hammer), incus (anvil) and stapes (stirrup)**?

- a. Hyaline cartilage
- b. Elastic cartilage
- c. Fibrocartilage
- d. All of the above
- e. None of the above

_____32. Which **connective tissue cell type** is functionally related to the **basophils**?

- a) Fibroblast
- b) Mast cell
- c) Histiocyte
- d) Plasma cell
- e) Myofibroblast

_____33. Which type of **connective tissue** is avascular and further characterized by ground substance in the form of a firm, resilient solid with a high proportion of glycosaminoglycans and with cells isolated in lacunae?

- a) adipose tissue
- b) cartilage
- c) tendon
- d) bone
- e) areolar tissue

_____34. With regard to **bone**, which statement is **false**?

- a) Compact (dense) bone is characterized by the presence of osteons.
- b) Bone formation without a pre-existing cartilage model is called intramembranous bone formation.
- c) In the restructuring of the skeleton, bone remodeling follows a cycle in which bone formation alternates with resorption.
- d) Osteoblasts secrete the components of bone matrix.
- e) Osteocytes are involved in bone resorption.

_____35. Which of the following statements is **correct** regarding the **bone**?

- a) Primary bone has a higher mineral content than the secondary bone.
- b) Sharpey's fibers are calcified collagen bundles that run in concentric manner.
- c) Bone contains matrix with more inorganic components than organic component in its dry weight.
- d) Osteoclasts have processes that run in their canaliculi.
- e) Osteoblasts are derived from blood monocytes.

_____36. Which of the following hormones is **most** responsible for increasing **osteoclast** number and activity?

- a) ADH
- b) calcitonin
- c) estrogen
- d) parathyroid hormone
- e) testosterone.

_____37. Select the **correct statement**.

- a) Endosteum is attached to the surface of bone by Sharpey's fibers.
- b) Compact (dense) bone is characterized by the presence of osteons.
- c) Dense regular connective tissue is abundantly found in the bone matrix.
- d) Osteocyte is involved in laying down osteoid.
- e) Howship's lacuna is unmineralized bone matrix.

_____38. Which of the following cells **do not** belong to the **mononucleated phagocytic system**?

- a) Microglia
- b) Dust cells
- c) Langerhan's cells
- d) Kupffer cells
- e) Hepatocytes

_____39. **Compact (dense) bone,**

- a) is the predominant bone tissue in the epiphyses of adult long bones.
- b) is characterized by the presence of osteons.
- c) is also called cancellous bone.
- d) can be either primary or secondary bone.
- e) is typically more radiolucent than other types of bones.

_____40. With regard to the **muscular system**, which statement is **false**?

- a) Z-lines represent attachment sites for thick filaments (myosin).
- b) Cardiac muscle consists of individual cells attached end-to-end by intercalated disks.
- c) The distance between two Z-lines in skeletal muscle is equivalent to the length of a sarcomere.
- d) The electrical signal of an action potential is conveyed into the interior of each muscle fiber by transverse tubules.
- e) Skeletal muscle can be described as syncytial, with many nuclei per "cell" located at the periphery of its muscle fibers, close to the fiber membrane.

_____41. The **skeletal muscle** contains

- a) caveolae
- b) intercalated disks
- c) triads
- d) no T tubules
- e) centrally located nuclei

_____42. With regard to the **skeletal muscle fibers**, which statement is **false**?

- a. Red fibers have more mitochondria than white fibers.
- b. Red fibers store glycogen and use anaerobic metabolism.
- c. Red fibers are more resistant to fatigue than are white fibers.
- d. Red fibers are smaller in diameter; white fibers are larger in diameter.
- e. Red fibers have more myoglobin (oxygen binding pigment) than white fibers.

_____43. What type of **muscle fiber** probably predominates in the legs of Haile Gebreselassie?

- a. Red fibers
- b. White fibers
- c. Intermediate fibers
- d. All of the above
- e. None of the above

_____44. What line bisects the **dark band in a striated muscle**?

- a. A band
- b. I band
- c. Z line
- d. H band
- e. M line

_____45. What region is made of **thin filaments**?

- a. A band
- b. I band
- c. Z line
- d. H band
- e. M line

_____46. Which of the following tissues is characterized by **abundant capillaries**?

- a) articular cartilage
- b) cardiac muscle
- c) dense regular connective tissue
- d) elastic cartilage
- e) stratified squamous epithelium

_____47. Which fiber type is seen in **smooth muscle**?

- a. Red fibers
- b. White fibers
- c. Intermediate fibers
- d. All of the above
- e. None of the above.

_____48. With regard to the **muscle tissue**, which statement is **false**?

- a. All types of muscle contain actin and myosin.
- b. Sarcolemma is the plasma membrane of a muscle cell.
- c. Both smooth muscle and cardiac muscle have sarcomeres.
- d. Intercalated disks consist of desmosomes, fascia adherens and gap junctions.
- e. A sarcomere of a skeletal muscle is the segment that runs from Z line to Z line.

_____49. The **electrical signal of an action potential** is conveyed into the interior of each muscle fiber by:

- a) Sarcomere
- b) transverse tubules
- c) sarcoplasmic reticulum
- d) Z line
- d) thick filaments

- _____50. Which of the following tissue types **does not** include **secretion** as a major function of its constituent cells?
a) connective tissue b) epithelial tissue c) muscle tissue d) nerve tissue e) all of them
- _____51. Which of the following cell types is found predominantly in **white matter**?
a) fibrous astrocytes b) microglia c) satellite cells d) protoplasmic astrocytes e) Schwann cells
- _____52. **Nissl bodies**,
a) are components of the RER that lack ribosomes. b) undergo chromatolysis after injury to the axon.
c) are found in the axon hillock. d) synthesize glycosaminoglycans. e) are PAS positive.
- _____53. With regard to the **dendrites**,
a) each has a constant diameter along its entire length. b) contain neurosecretory vesicles.
c) may contain Nissl bodies. d) each ends with a typical terminal bouton. e) are typically myelinated.
- _____54. Which of the following statements is **true** concerning the **nervous tissue**?
a) Schwann cells provide myelin for several axons simultaneously.
b) Unmyelinated axons generally have a smaller diameter than myelinated axons.
c) Action potentials normally travel from the axon terminal toward the cell body.
d) Gray matter in the central nervous system consists of axon tracts with abundant myelin.
e) Microglia line the central canal of the spinal cord and the ventricular system of the brain.
- _____55. A typical **peripheral mixed nerve** includes all of the following **EXCEPT**:
a) Schwann cells. b) astrocytes. c) sensory axons d) motor axons
e) connective tissue of the perineurium and/or endoneurium
- _____56. Which of the following statements is **not true** regarding the **supporting cells in the nervous tissue**:
a) Satellite cells surround the perikarya of the spinal ganglia in a stratified cuboidal fashion.
b) Oligodendrocytes are found in both gray and white matters of the central nervous system.
c) Astrocytes serve in conducting metabolites between blood vessels and neurones.
d) Microglia are the smallest in size amongst the neuroglial cells. e) None of the above.
- _____57. With respect to the **nervous tissue**, which of the following statements is **true**?
a) Action potentials travel fastest along the neuroglial cells than the neurons
b) Action potentials normally travel from the axon terminal toward the cell body.
c) Myelinated axons generally have a smaller diameter than unmyelinated axons
d) Dendrites are found predominantly in white matter e) None of the above.
- _____58. Which of the following statements is **not true** regarding the **supporting cells in the nervous tissue**:
a) Astrocytes serve in conducting metabolites between blood vessels and neurons.
b) Satellite cells surround the perikarya of the spinal ganglia in a stratified cuboidal fashion.
c) Oligodendrocytes are found in both gray and white matters of the central nervous system.
d) Microglia are the smallest in size among the neuroglial cells. e) None of the above.
- _____59. Which of the following cell types may provide **myelin for several axons simultaneously**?
a) ependymal cells b) oligodendrocytes c) microglia d) Schwann cells e) satellite cells

_____60. **Pericytes** are

- a) specialized cardiac muscle cells
- b) are attached to the outside of capillaries
- c) specialized smooth muscle cells
- d) terminally differentiated
- e) multinucleated

_____61. A differential cell count of a **blood smear** from a patient with a parasitic infection is likely to reveal an increase in the circulating numbers of which of the following blood cell types?

- a) basophils
- b) eosinophils
- c) neutrophils
- d) monocytes
- e) platelets

_____62. Which of the following cell types is capable of **further mitosis after leaving the hematopoietic organ** in which they were formed and are capable of returning to the **circulation after leaving the blood** to enter the connective tissue?

- a) basophils
- b) eosinophils
- c) lymphocytes
- d) monocytes
- e) platelets

_____63. The order of frequency of the leukocytes is:

- a) neutrophil, lymphocyte, monocyte, eosinophil, basophil.
- b) neutrophil, monocyte, lymphocyte, eosinophil, basophil.
- c) neutrophil, eosinophil, lymphocyte, monocyte, basophil.
- d) neutrophil, lymphocyte, platelets, eosinophil, basophil.
- e) neutrophil, lymphocyte, basophil, monocyte, eosinophil.

_____64. Which of the following cell types is most likely to be **absent** in a differential cell count of **blood smear** from a normal patient?

- a) basophils
- b) eosinophils
- c) erythrocytes
- d) lymphocytes
- e) neutrophils

_____65. **Megakaryocytes,**

- a) are multinucleated
- b) are formed by the fusion of many haploid cells
- c) serve as precursors to bone marrow macrophages
- d) are located primarily in the spleen
- e) contain a network of intracellular membranes.

_____66. Which of the following **agranulocyte** cell types **becomes phagocytic after it enters the connective tissues?**

- a) eosinophils
- b) erythrocytes
- c) lymphocytes
- d) neutrophils
- e) monocytes

_____67. Which of the following cell types has cytoplasmic granules that contain **heparin and histamine?**

- a) eosinophils
- b) basophils
- c) lymphocytes
- d) neutrophils
- e) monocytes

_____68. Which of the following substances is primarily responsible for **maintaining the osmotic pressure of blood?**

- a) albumin
- b) fibrinogen
- c) immunoglobulin
- d) glucose
- e) serotonin

_____69. Which cell type has a nucleus with a checker-board pattern and grayish-staining cytoplasm because of the presence of roughly equal amount of basophilic and acidophilic components?

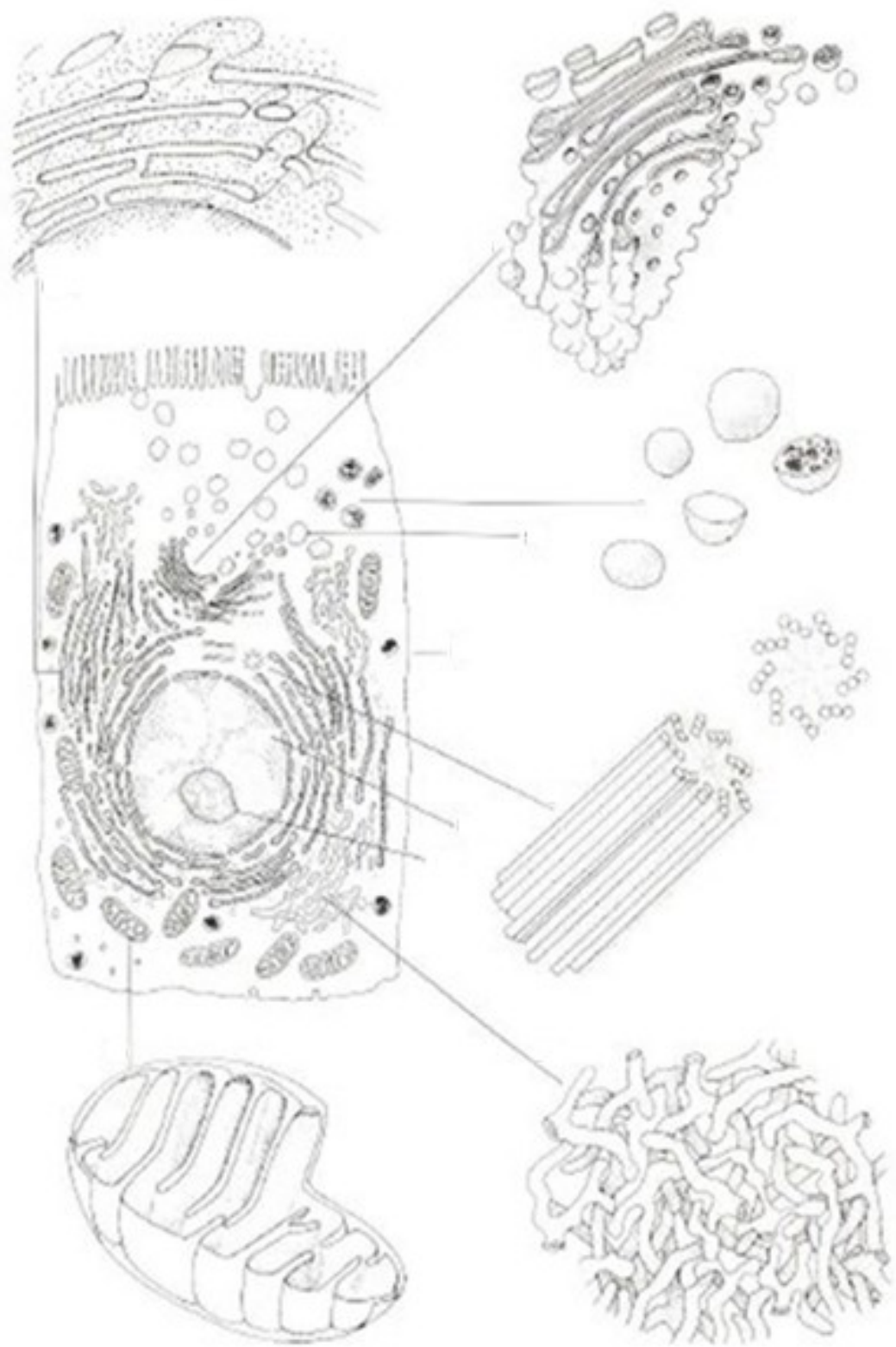
- a) basophilic erythroblast
- b) orthochromatophilic erythroblast (normoblast)
- c) proerythroblast
- d) polerythroblast
- e) reticulocyte

_____70. Which of the following is the earliest stage at which **azurophilic granules** first accumulate?

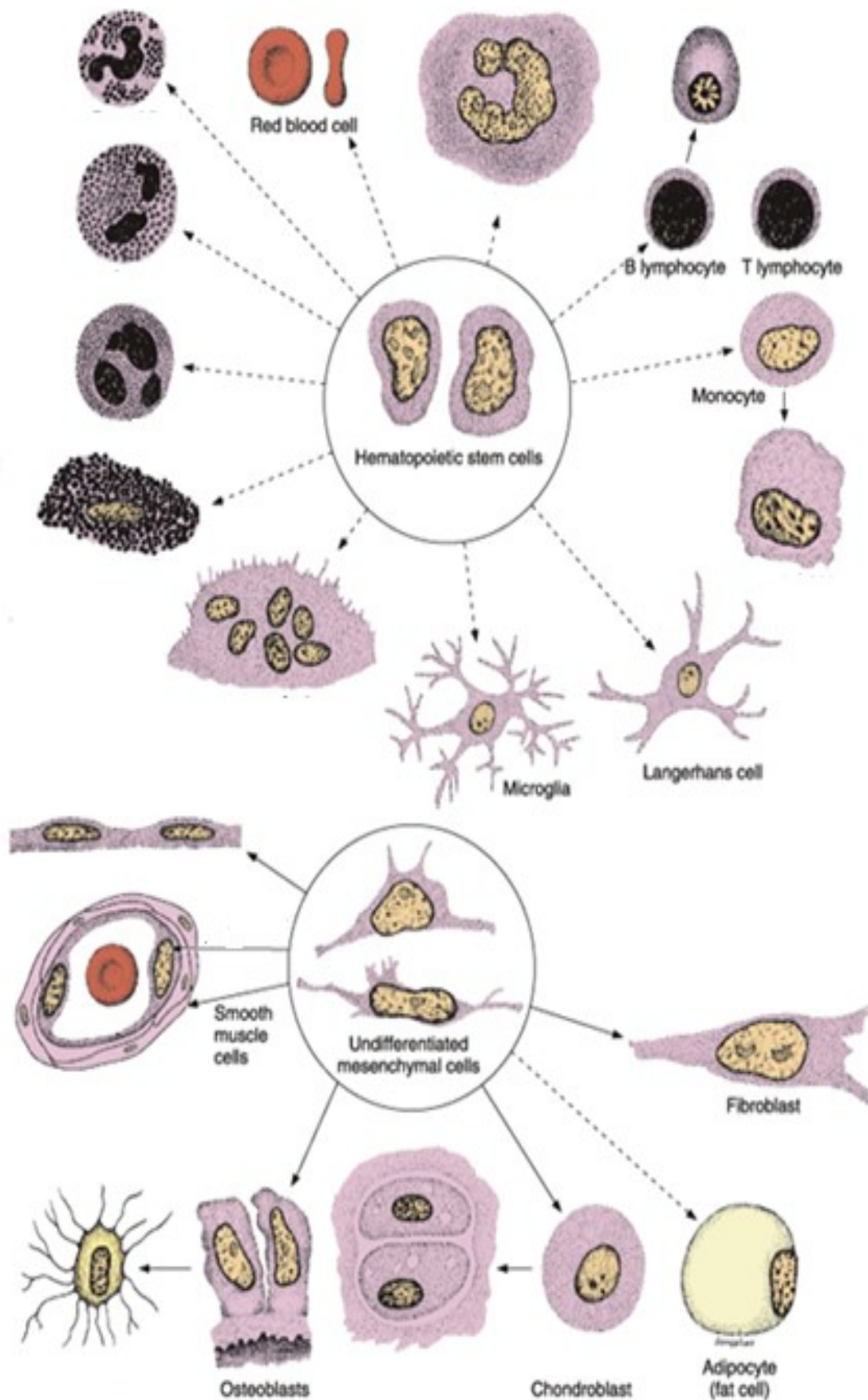
- a) band form
- b) metamyelocyte
- c) myeloblast
- d) myelocyte
- e) promyelocyte

PART II. Label the following diagrams.

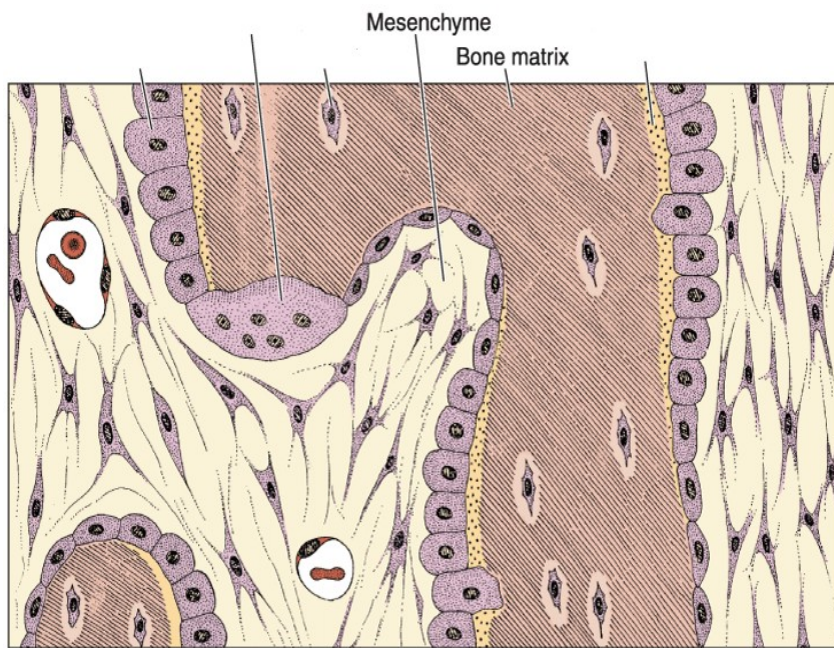
1. Hypothetical cell



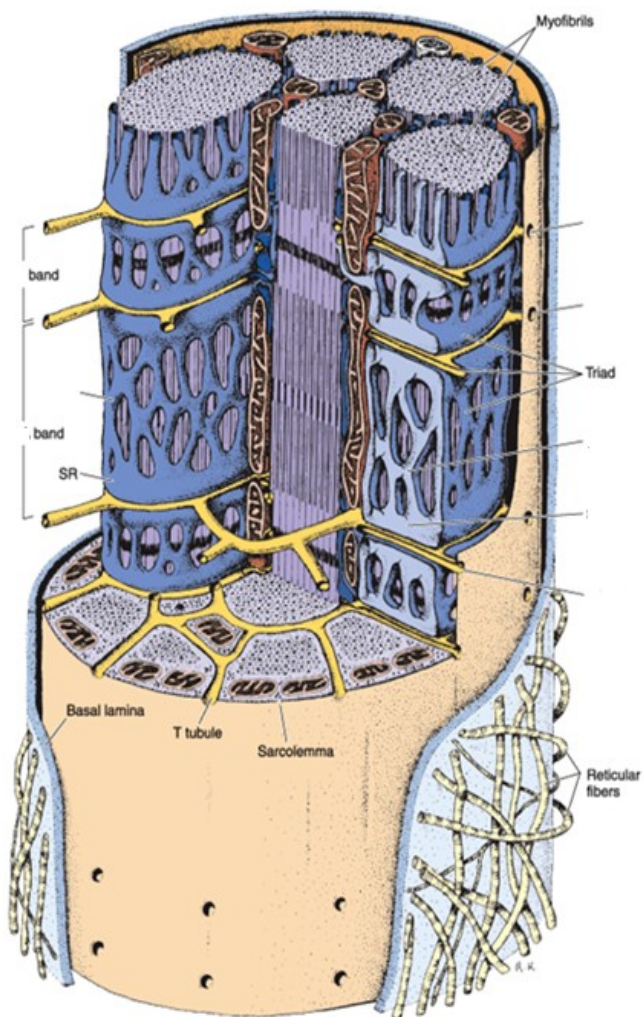
2. Simplified representation of the ConnectiveTissue cell lineage.



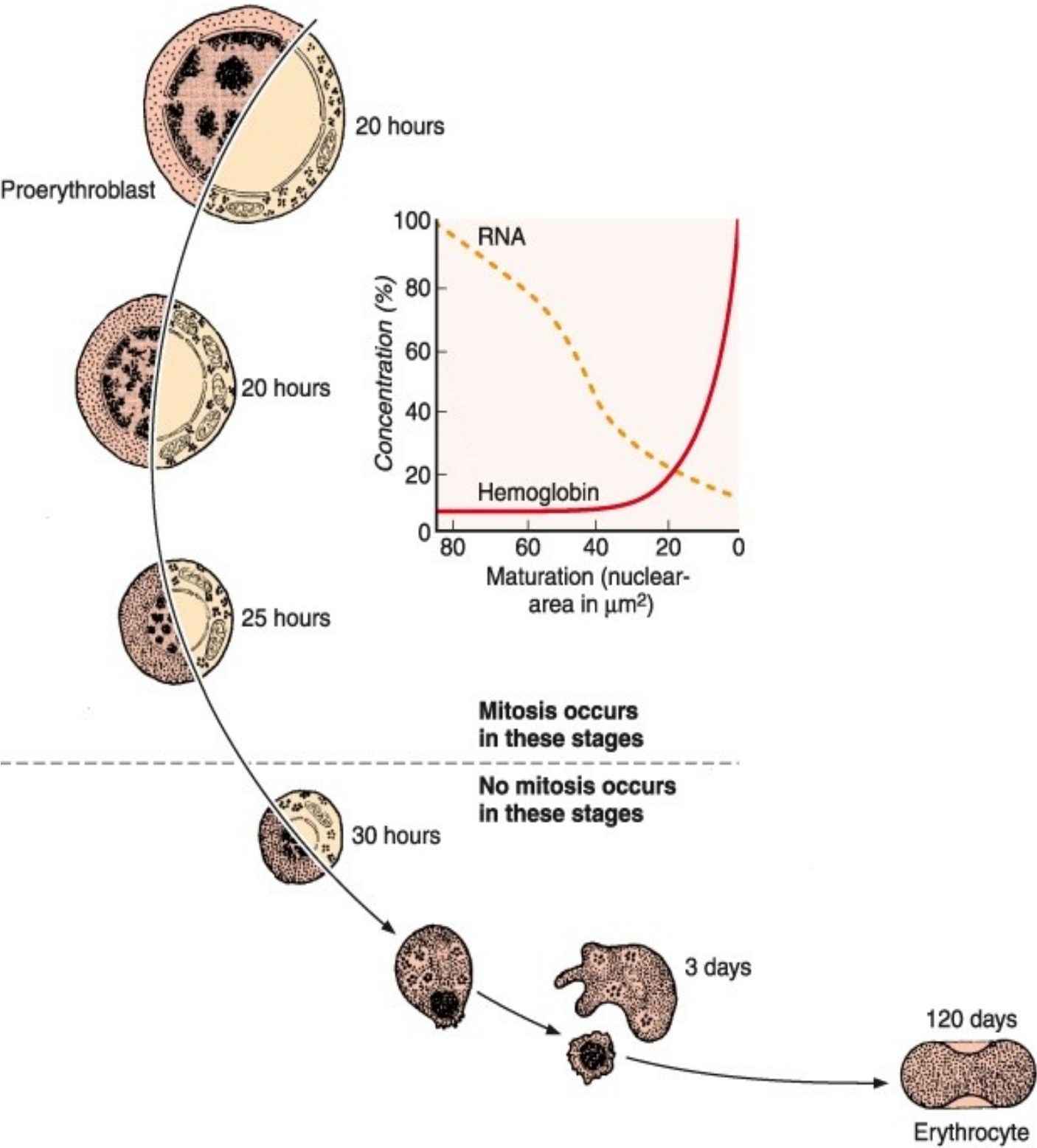
3. Events that occur during intramembranous ossification.



4. Segment of skeletal muscle.



5. Summary of erythrocyte maturation.



PART III. Answer the following questions.

1. *Describe the cell cycle.*
2. *Enumerate the functions of epithelial tissue.*
3. *Enumerate the functions of connective tissue.*
4. *What roles do vitamins and hormones play in the growth of bones?*
Emphasis: excess and deficiency.
5. *What are the components of the blood plasma? Describe the functions of each component.*

DEFENSE MEDICAL COLLEGE

Progressive Test I in Embryology for Year I Medical Students

Time allowed: 2:00 hours.

Date: March 12, 2014

Name: _____

Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer that is best in each.

- _____ 1. Concerning ***oogenesis***, which statement is **true**?
- a) All oogonia become primary oocytes during ovulation.
 - b) Primary oocytes begin the first meiotic division before birth.
 - c) Oogonia grow and give rise to secondary oocytes by mitotic division.
 - d) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
 - e) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.
- _____ 2. With regard to ***oogenesis***, which statement is **false**?
- a) Primary oocytes remain dormant in prophase of meiosis I until puberty.
 - b) If fertilization does not occur, the secondary oocyte becomes primary oocyte.
 - c) Primary oocytes remain arrested in metaphase of mitosis II until fertilization occurs.
 - d) All primary oocytes are formed postnatally.
 - e) None of the above.
- _____ 3. When does a ***secondary oocyte*** complete its second meiotic division to become a mature ovum?
- a) at ovulation
 - b) before ovulation
 - c) at fertilization
 - d) at puberty
 - e) before birth
- _____ 4. Concerning ***spermatogenesis***, one statement is **false**.
- a) Type A spermatogonia undergo mitosis to provide a continuous supply of stem cells throughout the reproductive life of the male.
 - b) The primary spermatocytes are the largest germ cells in the seminiferous tubules.
 - c) Primary spermatocytes in males begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.
 - d) No primary spermatocytes form after birth.
 - e) None of the above.
- _____ 5. With regard to ***spermatogenesis***, which statement is **false**?
- a. spermatids are transformed into spermatozoa by spermiogenesis.
 - b. The entire process of spermatogenesis takes approximately 72 hours.
 - c. Spermatogenesis is regulated by LH production by the pituitary.
 - d. Spermatogenesis begins at puberty.
 - e. Spermatogonia begin to increase in number at puberty.
- _____ 6. Regarding the ***sperm transport***, which statement is **false**?
- a) Vesiculase (an enzyme in the seminal vesicle) coagulates some of the semen and forms a vaginal plug.
 - b) Fructose in the semen, secreted by the seminal glands, is an energy source for the sperms.
 - c) Spermatozoa are nonmotile during storage in the epididymis, but become motile in the ejaculate.
 - d) Prostaglandins in the semen stimulate uterine motility.
 - e) None of the above.

- _____ 7. With regard to **Sertoli cells**, which statement is **false**?
- a. They line the seminiferous tubules
 - b. They support and nurture the male germ cells.
 - c. They allow random assortment of maternal and paternal chromosomes between the gametes.
 - d. They are involved in the regulation of spermatogenesis.
 - e. They produce testicular fluid.
- _____ 8. Which statement is **true** about **female reproductive cycles**?
- a) Ovulation is triggered by a surge of FSH production.
 - b) If an oocyte is fertilized, the corpus luteum involutes and degenerates.
 - c) LH stimulates the development of ovarian follicles and the production of estrogen by its follicular cells.
 - d) Ischemic phase occurs when the oocyte is fertilized.
 - e) None of the above.
- _____ 9. Concerning the **female reproductive system**,
- a) The usual site of fertilization is the ampulla of the uterine tube.
 - b) The endometrium has a wall made up of a very thick smooth muscle layer.
 - c) The basal layer of endometrium is sloughed off during each menstruation.
 - d) The perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
 - e) The vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.
- _____ 10. Which statement is **false**?
- a. If a sperm penetrates the secondary oocyte, the second meiotic division is completed.
 - b. The number of oocytes that ovulate is greatly reduced in women who take oral contraceptives.
 - c. Removal of the corpus luteum of pregnancy before the 4th month usually leads to abortion.
 - d. At ovulation, the nucleus of the secondary oocyte begins the second meiotic division, but progresses only to metaphase.
 - e. None of the above.
- _____ 11. With respect to the **zona pellucida**, which statement is **false**?
- a. is composed of sulfated glycoproteins.
 - b. arises by delamination from the embryoblast.
 - c. prevents polyspermy.
 - d. prevents premature implantation.
 - e. functions as a porous filter.
- _____ 12. **Liquor folliculi**,
- a) stimulates the myoepithelial cells of the mature follicle to rupture.
 - b) causes the endometrial glands to secrete and prepare the endometrium for implantation.
 - c) is composed of cumulus oophorus and polar bodies.
 - d) acts as a porous filter through which dissolved hormones and enzymes are prevented from entering the primary follicle.
 - e) None of the above.
- _____ 13. Which statement is **false** about **fertilization**?
- a) It restores the normal diploid number of chromosomes.
 - b) It stimulates the penetrated oocyte to complete the first meiotic division.
 - c) It causes metabolic activation of the ovum and initiates cleavage of the zygote.
 - d) It results in variation of the human species.
 - e) It determines the chromosomal sex of the embryo.
- _____ 14. If **fertilization does not occur**:
- a. The luteal phase continues and menstruation does not occur.
 - b. The blastocyst begins to implant in the endometrium on exactly the 6th day of the luteal phase.
 - c. Human chorionic gonadotropin keeps the corpus luteum and keep on secreting estrogens and progesterone.
 - d. Syncytiotrophoblast manufactures most of the body's progesterone in females.
 - e. None of the above.

- _____ 15. If **fertilization** occurs:
- a) The corpus luteum degenerates.
 - b) Estrogen and progesterone levels fall.
 - c) Menstruation occurs.
 - d) Cleavage of the zygote and blastogenesis occur.
 - e) The secretory endometrium enters an ischemic phase.
- _____ 16. Before a **spermatozoan fertilizes an ovum**.
- a) Its glycoprotein coat must be removed from its head.
 - b) The zona pellucida must be shed.
 - c) Its acrosomal membrane must be impermeable.
 - d) Its tail piece should be lost.
 - e) Its should be capacitated in the epididymidis.
- _____ 17. Regarding the **phases of fertilization**, which statement is **false**?
- a) Passage of sperm through corona radiata.
 - b) Penetration of zona pellucida.
 - c) Fusion of the male and female pronuclei into a single diploid aggregation of chromosomes, the ovum becomes a zygote.
 - d) Fusion of plasma membranes of the oocyte and sperm.
 - e) Completion of the first meiotic division of spermatozoa and formation of the male pronucleus.
- _____ 18. What is the result of the cortical reaction?
- a) The corona radiata forms
 - b) The zona pellucida forms
 - c) Sperm can now enter the ovum
 - d) Sperm can no longer enter the ovum
 - e) Acrosomes are removed
- _____ 19. **Development of an ovarian follicle** is not characterized by:
- a. Growth and differentiation of primary oocyte
 - b. Proliferation of follicular cells
 - c. Formation of zona pellucida
 - d. Development of the theca folliculi
 - e. Inhibition of primordial follicles by interstitial cells of Leydig.
- _____ 20. Which of the following componenets plays the most active role in **invading the endometrium** during **blastocyst implantation**?
- a) Epiblast
 - b) Syncytiotrophoblast
 - c) Hydroblast
 - d) Extraembryonic somatic mesoderm
 - e) Extraembryonic visceral mesoderm
- _____ 21. Which of the following structures must degenerate for **blastocyst implantation** to occur?
- a) endometrium in progestational phase
 - b) zona pellucida
 - c) syncytiotrophoblast
 - d) cytotrophoblast
 - e) functional layer of the endometrium.
- _____ 22. Where does the **blastocyst normally implant**?
- a) functional layer of the cervix
 - b) myometrium
 - c) functional layer of the endometrium
 - d) perimetrium
 - e) basal layer of the endometrium
- _____ 23. During the **first week of development**,
- a) Somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
 - b) The embryo is a solid ball of blastomeres and is called gastrula.
 - c) Amniotic cavity is formed.
 - d) Neural tube is formed.
 - e) None of the above.
- _____ 24. During the **first week of development**, which statement is **false**?
- a) the embryoblast, which gives rise to the embryo and some extraembryonic tissues, is formed.
 - b) the trophoblast at the embryonic pole differentiates into cytotrophoblast and syncytiotrophoblast.
 - c) gastrulation takes place.
 - d) a blastocyst cavity is formed.
 - e) None of the above.

25. Which of the following is **not** a **developmental event of the 2nd week**?
- a) Formation of amniotic cavity b) Formation of the notochord c) Formation of chorion
d) Completion of implantation e) Formation of bilaminar germ disc
26. During the **second week of development**, which statement is **false**?
- a) The end of the second week is characterized by the appearance of primary chorionic villi.
b) The epiblast forms the floor of the amniotic cavity and is continuous peripherally with the amnion.
c) Enough hCG is produced by the syncytiotrophoblast at the end of the second week to give a positive pregnancy test.
d) The 10-day human conceptus (embryo and extraembryonic membranes) is completely embedded in the endometrium.
e) None of the above.
27. On day nine, trophoblastic lacunae and maternal sinusoids begin to form around what cells:
- a) Exdoderm b) Extraembryonic coelom d) Exocoelomic
c) Extraembryonic splanchnopleuric e) Syncytiotrophoblast
28. A patient is being seen due to hypertension and vaginal bleeding, four months into her pregnancy. High levels of hCG are found but no inner cell mass (embryo) is present. What is likely her diagnosis?
- a) Hydatidiform mole b) Ectopic pregnancy c) Placenta previa
d) Placenta abruption e) Placenta accrete
29. Regarding **cleavage**, which statement is **false**?
- a) Results in the formation of increasingly small cells.
b) Occurs as the zygote passes down the uterine tube.
c) Begins when the male and female pronuclei contact each other.
d) Enhances the corpus luteum to enlarge to form corpus albicans.
e) Is a series of rapid mitotic divisions while the zygote is still surrounded by the zona pellucida.
30. Select the **incorrect** statement.
- a) Gastrulation is first indicated by the formation of the primitive streak.
b) Neurulation is the process by which neuroectoderm forms the neural plate.
c) Preponderance of sacrococcygeal teratoma is found in male newborns. (very common)
d) All germ layers are derived from epiblast. e) Epiblast and hypoblast are fused in the cloacal membrane.
31. Which statement is **false** about **prochordal plate**?
- a) Helps to establish the bilateral symmetry of the embryo.
b) Helps to establish the cranial versus the caudal region of the embryo.
c) It eventually develops into oropharyngeal membrane at the site of the future mouth.
d) It is columnar-shaped thickening of epiblast cells. e) It is a circular, midline thickening of hypoblast cells.
32. With regard to the **notochord**, which statement is **false**?
- a) It forms the nucleus pulposus of each intervertebral disk.
b) It is a solid cylinder of mesoderm extending from the primitive node to the prochordal plate.
c) Provides signals that are necessary for the development the central nervous system.
d) It induces the formation of cardiogenic region. e) It induces the formation of each vertebral body.
33. Which of the following is/are **not mesodermal in origin**?
- a) Astrocytes b) Bone and cartilage c) Microglia
d) Cardiac muscle cells e) Endothelium of blood and lymph vessels

- _____ 34. With regard to the **primitive streak**, select the statement which is **false**.
- a) It forms the nucleus pulposus of each intervertebral disk.
 - b) Actively forms mesoderm until the early part of the fourth week.
 - c) Results from migration of epiblastic cells to the median plane of the disc.
 - d) Is a thickening of the epiblast at the caudal end of the embryonic disc at the beginning of the 3rd week.
 - e) Normally undergoes degenerative changes but remnants may persist and give rise to sacrococcygeal teratoma.
- _____ 35. Which of the following is/are **not** a derivative(s) of the **neural crest**?
- a) Schwann cells b) Chromaffin cells c) Neurohypophysis d) Melanocytes e) Parafollicular (C) cells
- _____ 36. **Intermediate mesoderm** gives rise to the
- a) Neural tube b) Heart c) Kidneys and gonads d) Somites e) Notochord
- _____ 37. Which germ layer is responsible in giving rise to the **epithelial tissue**?
- a) Ectoderm b) Mesoderm c) Endoderm d) a & c only e) All of them
- _____ 38. What germ layer forms the **melanocytes**?
- a) Endoderm _____ b) Neural crest c) Paraxial mesoderm
 - d) Intermediate mesoderm e) Lateral plate mesoderm
- _____ 39. Which structure is/are **not** visible during the **fourth week**?
- a) closed neuropores b) three pairs of pharyngeal arches c) scalp vascular plexus
 - d) upper limb buds e) lens placodes.
- _____ 40. During the **embryonic period**,
- a) Pigment is visible in the retina b) Sex of the embryo is clearly distinguishable externally
 - c) Ossification of the fetal skeleton is active. d) The amount of white fat is about 16% of body weight.
 - e) The body of the embryo is usually completely covered with lanugo hairs and vernix caseosa.
- _____ 41. Select the **correct** event during the **fetal period**.
- a) Fetal allantois is an important site of erythropoiesis during week 20.
 - b) The quantity of white fat is about 8% of body weight during week 21.
 - c) By 36 weeks, the circumferences of the head and abdomen are approximately equal.
 - d) The eyelids open at week 18. e) Umbilical herniation is a normal event during the 38th week.
- _____ 42. Concerning the **placenta**, which statement is **false**?
- b) Fetal part develops from the chorionic sac (villous chorion).
 - c) Trophoblasts are the precursor cells of the human placenta.
 - e) Is made up of the chorion frondosum together with the decidua basalis.
 - d) Cotyledons are partially separated by the 3 cm- thick chorionic plate and amnion.
 - a) It typically is discoid, hemochorial and weighs approximately 500-600 grams at term.
- _____ 43. Concerning the **fetal membranes (developmental adnexa)**, which statement is **true**?
- a) The yolk sac (umbilical vesicle) helps control the embryo's body temperature.
 - b) As the bladder enlarges, the umbilical cord involutes to form urachus.
 - c) Blood formation occurs in the wall of allantois during the third to fifth weeks.
 - d) Primordial germ cells appear in the endodermal lining of the wall of the tertiary chorion.
 - e) During the 4th week the mesoderm of the yolk sac is incorporated into the embryo as the primitive streak.

44. Which of the following does **NOT** function in hemopoiesis?

- a) Yolk sac b) Liver c) Heart d) Bone Marrow e) Spleen

45. Which of the following condition is **not** seen in ***prolonged pregnancy***?

- a) No lanugo hairs b) Increased vernix caseosa c) Long nails
d) Over weight e) Increased alertness

46. With regard to the **placenta and fetal membranes**, which statement is **false**?

- a) The finger-like projections of cytotrophoblast into syncytiotrophoblast are called primary chorionic villi.
b) At the fetomaternal junction, projections of decidua basalis towards the chorionic plate serve to divide the fetal placenta into irregular areas called anchoring villi.
c) Blood in the intervillous space is considered outside the mother's circulatory system and adequate irrigation of chorionic villi is an important factor in the development of the fetus.
d) Normally there are two arteries and one vein in the the umbilical cord.
e) Allantois is responsible for bladder development.

47. With regard to the **placenta**, which statement is **true**?

- a) Deoxygenation of blood is one of the major functions of the placenta.
b) As pregnancy advances, the placental membrane becomes progressively thicker.
c) The fetal part of the placenta is attached to the maternal part of the placenta by the cytotrophoblastic shell.
d) The maternal part of the placenta develops from the chorionic sac.
e) Amniochorionic membrane ruptures during labor.

48. Which of the following statements is **false**?

- a) The amniotic sac enlarges faster than the chorionic sac.
b) The amnion and smooth chorion fuse to form the amniochorionic membrane.
c) The placenta during early pregnancy synthesizes glycogen, cholesterol, and fatty acids.
d) High levels of alpha fetoprotein usually indicate the presence of a severe neural tube defect.
e) As pregnancy advances, the placental membrane becomes progressively thicker and, therefore, acts as a perfect barrier to infections.

49. Concerning **multiple pregnancies**,

- a) Twins with one amnion, one chorion and one placenta are always monozygotic.
b) The frequency of monozygotic twinning shows marked racial differences.
c) The rate of dizygotic twinning increases with maternal age.
d) Dizygotic twins always have two amnions and two chorions. e) None of the above.

50. Select the **correct statement**.

- a) The two layers of extraembryonic mesoderm are called epiblast and hypoblast.
b) The decidua reaction protects the uterus (endometrium) from implantation.
c) Primary chorionic villi are finger-like projections of cytotrophoblast into syncytiotrophoblast.
d) The axial skeleton forms from the paraxial mesoderm. e) None of the above.

PART II. Answer the following questions.

1. How would you reasonably estimate the age of an embryo?
2. What are the causes of Intrauterine growth restriction (IUGR).
3. Enumerate the functions of the amniotic fluid.